



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2012
YEAR 11 Mathematics
Yearly

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answer must be given in simplest exact form.

Total Marks – 70

- Attempt questions 1-15

Examiner: *P. Bigelow*

Section I (10 marks)

Answer this section on the Multiple Choice Answer Sheet

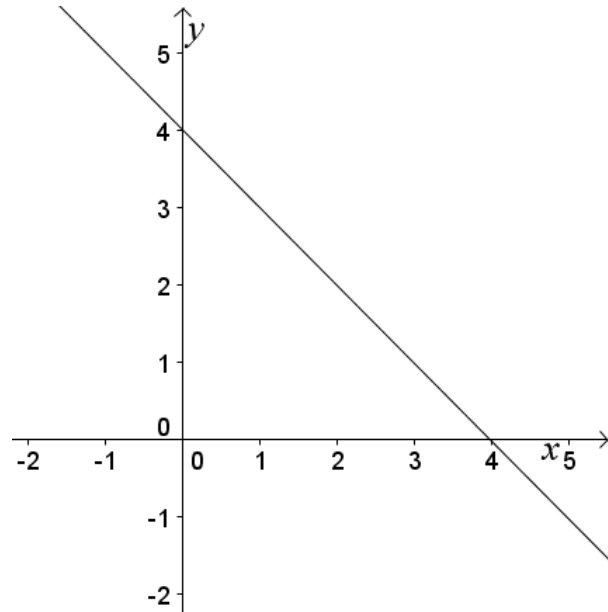
(1) The equation of the line is:

(A) $x - y + 4 = 0$

(B) $x + y - 4 = 0$

(C) $y = -x - 4$

(D) $x - y - 4 = 0$



(2) $x^2 - 4x + 6$ has a minimum value of:

(A) 4

(B) 2

(C) 6

(D) 8

(3) $3^x \times 2^x$ is equal to:

(A) 6^x

(B) 6^{2x}

(C) 5^x

(D) 5^{2x}

(4) The full solution to $x^2 = 2x$ is:

- (A) $x = 2$
- (B) $x = -2, 0$
- (C) $x = 0, 2$
- (D) $x = -2$

(5) The solution to $\log_5 x = 4$ is:

- (A) 625
- (B) 25
- (C) 125
- (D) 20

(6) If $g(x) = 6 - x - x^2$ then $g(-2)$ is:

- (A) 6
- (B) 4
- (C) 0
- (D) 12

(7) If $25^{2-x} = 125$ then x is equal to:

- (A) 1
- (B) $\frac{1}{2}$
- (C) -1
- (D) 5

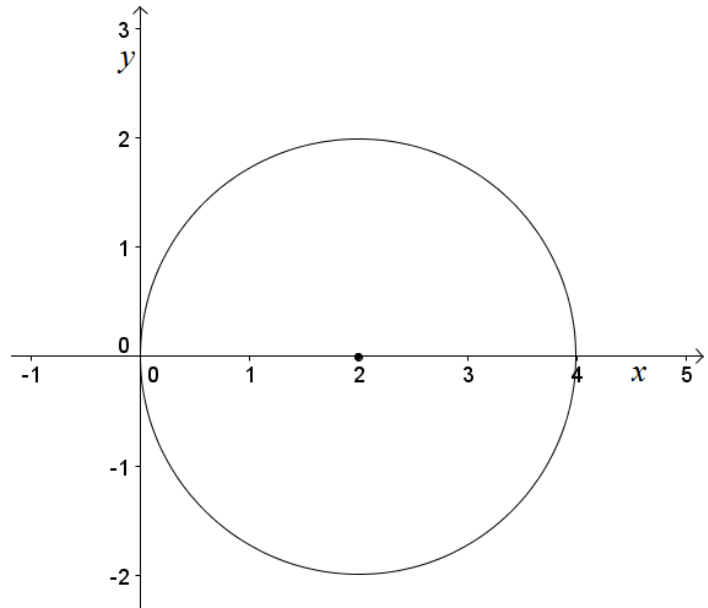
(8) The equation of the circle is:

(A) $x^2 + (y - 2)^2 = 4$

(B) $(x - 2)^2 + y^2 = 16$

(C) $(x + 2)^2 + y^2 = 16$

(D) $(x - 2)^2 + y^2 = 4$



(9) The domain for the function $f(x) = \sqrt{2 - x}$ is:

(A) $x \geq 0$

(B) $x \geq 2$

(C) $x \leq 2$

(D) All reals

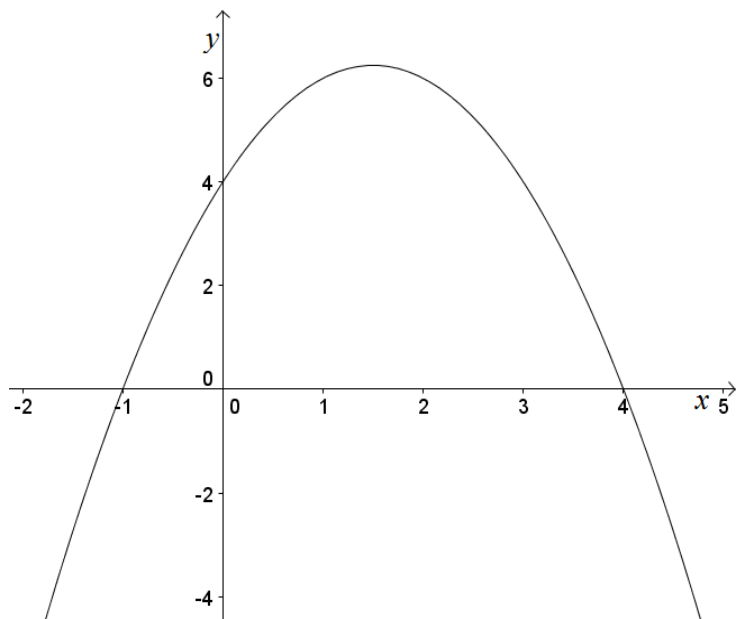
(10) The graph best represents

(A) $y = (x + 1)(x - 4)$

(B) $y = (x - 1)(x + 4)$

(C) $y = (1 - x)(x - 4)$

(D) $y = (x + 1)(4 - x)$



Section II (60 marks)

Answer this section in the Writing Booklets

Question 11 [11 marks]

(a) Sketch the following on separate number plane graphs [3]

(i) $x^2 + y^2 = 25$

(ii) $y = 4^x$

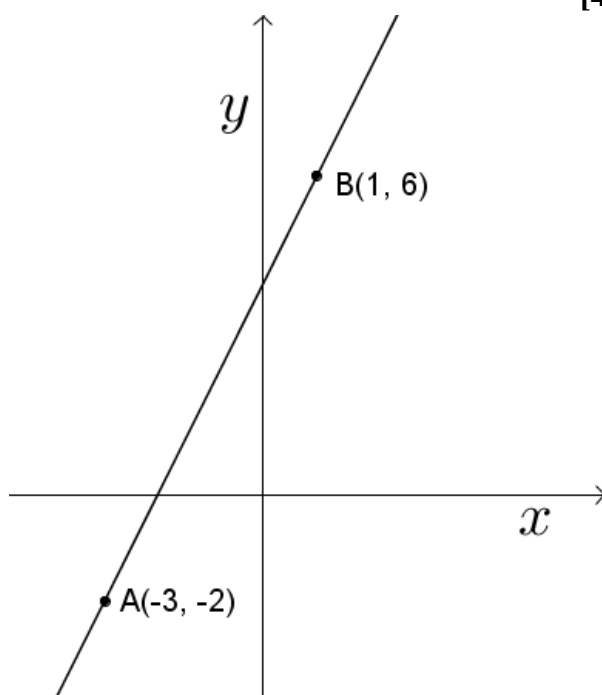
(iii) $xy \leq 3$

(b) Find [4]

(i) Mid-point of AB.

(ii) The gradient of AB.

(iii) The equation of AB.



(c) Find the gradient of the line $2x + 3y - 11 = 0$ [1]

(d) The point $(-3, c)$ lies on $x + 4y - 9 = 0$, find the value of c . [1]

(e) Find the perpendicular distance of the line $3x - 4y - 10 = 0$ from the origin. [2]

Question 12 [12 marks]

(a) Find the value of x in the following: [5]

(i) $\log_x 36 = 2$

(ii) $x = \log_3 \frac{1}{9}$

(iii) $\log_4 x = 3$

(iv) $2^{2x} = 16^{x-8}$

(b) Sketch $y = \log_2 x$ for $0 < x \leq 8$. [2]

(c) If $f(x) = x^2 - x$, find the values of a such that $f(a) = f(2)$. [2]

(d) State whether the following functions are even (E), odd (O) or neither (N). [3]

(i) $f(x) = 2 - x^2$

(ii) $f(x) = \frac{x}{1+x}$

(iii) $f(x) = x\sqrt{x^2 + 1}$

Question 13 [12 marks]

(a) Find the solutions to the following quadratic equations (leave in simplest exact form).

[4]

(i) $x^2 + 6x - 4 = 0$

(ii) $6x^2 - 17x = 14$

(b) Find the centre and radius of the circle $x^2 - 6x + y^2 + 8y = 0$.

[2]

(c) Find

[6]

(i) Focus

(ii) Directrix

(iii) Vertex

for the following

(α) $x^2 = 12y$

(β) $x^2 - 4x + 8y = 0$

Question 14 [11 marks]

(a) Find A, B and C such that

[3]

$$A(x + 1)^2 + B(x + 1) + C \equiv 3x^2 + 4x + 5$$

(b) Solve the following simultaneous equations

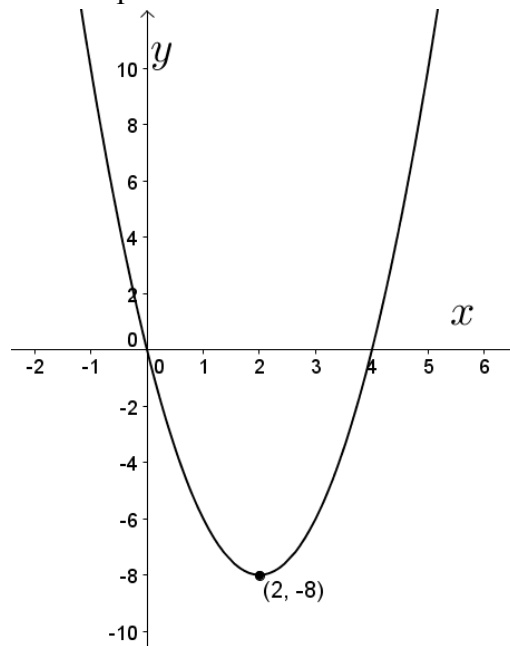
[3]

$$5^{x+y} = \frac{1}{5}$$

$$5^{3x+2y} = 1$$

(c) Find the equation of the parabola.

[2]



(d) Find the equation of the line perpendicular to $2x + y - 5 = 0$ and passing through $(5, 1)$.

[3]

Question 15 [14 marks]

(a) If the roots of $x^2 - 2x - 4 = 0$ are α and β find the value of: [5]

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iv) $\alpha^2 + \beta^2$

(b) For what values of k does $x^2 - (k + 5)x + 9 = 0$ have [3]

(i) Equal roots

(ii) No real roots

(c) Form a quadratic equation with roots $2 + \sqrt{3}$ and $2 - \sqrt{3}$. [2]

(d) The cost of producing y items per day is $\$ \left(\frac{1}{3}y^2 + 45y + 27 \right)$. The price at which each item may be sold is $\$ \left(60 - \frac{1}{2}y \right)$. [4]

(i) Find an expression for the daily profit.

(ii) Find the maximum value of the profit.

End of Exam



Student Number: Answers

Mathematics

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

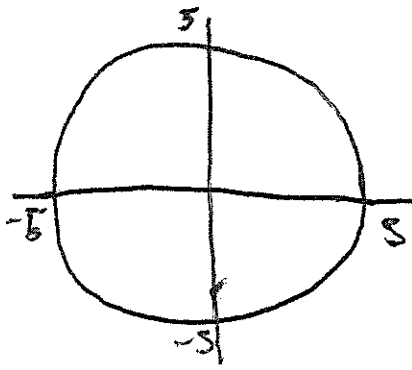
A ☒ B ☒ C ☐ D ☐
correct

Section I: Multiple choice answer sheet.

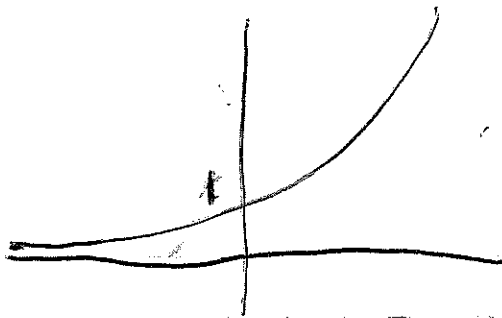
Completely colour the cell representing your answer. Use black pen.

1. A ☒ B ☐ C ☐ D ☐
2. A ☒ B ☐ C ☐ D ☐
3. A ☒ B ☐ C ☐ D ☐
4. A ☐ B ☒ C ☐ D ☐
5. A ☒ B ☐ C ☐ D ☐
6. A ☐ B ☒ C ☐ D ☐
7. A ☐ B ☒ C ☐ D ☐
8. A ☐ B ☒ C ☐ D ☐
9. A ☐ B ☒ C ☐ D ☐
10. A ☐ B ☐ C ☒ D ☐

Q11 (a)
(i)

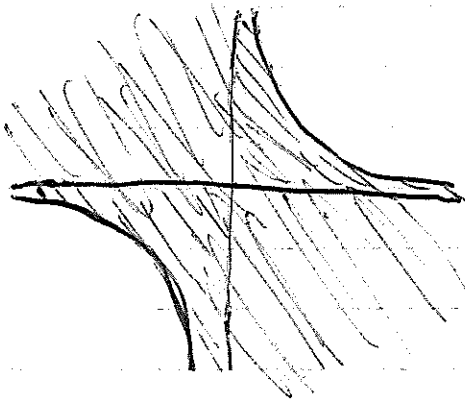


(ii)



3

(iii)



(b) (i) Mid Point $AB = \left(\frac{1+3}{2}, \frac{6+2}{2} \right)$
 $= (-1, 2)$

(ii) $m_{AB} = \frac{6+2}{1+3}$

4

$= 2.$

(iii) $y+2 = 2(x+3)$

$y = 2x + 4.$

$$(c) \quad 2x + 3y - 11 = 0$$

$$y = -\frac{2}{3}x + \frac{11}{3}$$

$$m = -\frac{2}{3}$$

$$(d) \quad -3 + 4c - 9 = 0$$

$$4c = 12$$

$$c = 3$$

(e)

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|-10|}{\sqrt{9+16}}$$

$$= 2 \text{ units}$$

$$Q12 (a) (i) \quad x = 6$$

$$(ii) \quad x = -2$$

$$(iii) \quad x \cdot 4^3 = 64$$

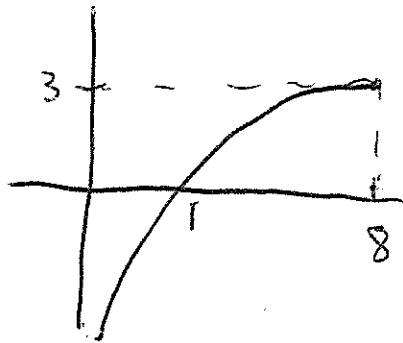
$$(iv) \quad 2^{2x} = 2^{4x-32}$$

$$2x = 4x - 32$$

$$2x = 32$$

$$x = 16$$

(b)



2

(c) $a^2 - a = 4 - 2$

$$a^2 - a - 2 = 0.$$

~~$a = \frac{1 \pm \sqrt{1+8}}{2}$~~

2

$$(a-2)(a+1) = 0$$

$$a = 2, -1.$$

- (d) (i) Even
(ii) Neither
(iii) Odd.

3

Q/3 (a) (i) $x^2 + 6x - 4 = 0.$

$$x = \frac{-6 \pm \sqrt{36 + 16}}{2} = \frac{-6 \pm \sqrt{52}}{2}.$$

~~$= \frac{-6 \pm 2\sqrt{13}}{2} = -3 \pm \sqrt{13}.$~~

~~$= -3 \pm \sqrt{13}.$~~

(ii) $6x^2 - 17x - 14 = 0$.

$$x = \frac{17 \pm \sqrt{17^2 - 4 \times 6 \times -14}}{12}$$

$$= \frac{17 \pm 25}{12}$$

$$= \frac{7}{2}, -\frac{2}{3} \quad 2$$

(b) $(x-3)^2 - 9 + (y+4)^2 - 16 = 0$

$$(x-3)^2 + (y+4)^2 = 25$$

centre $(3, -4)$
radius 5.

2

(c) (i) $(0, 3)$

(ii) $y = -3$

(iii) $(0, 0)$

3

(d) $(x-2)^2 = -8y + 4$

$$(x-2)^2 = -4 \times 2 (y - \frac{1}{2})$$

(i) $(2, -\frac{1}{2})$

(ii) $y = \frac{1}{2}$

(iii) $(2, \frac{1}{2})$

3

Q14
(a)

$$A(x+1)^2 + B(x+1) + C \equiv 3x^2 + 4x + 5,$$

when $x = -1$.

$$C = 3 - 4 + 5 \\ = 4$$

$$\underline{x^2} \quad A = 3.$$

constant.

$$A + B + C = 5$$

$$3 + B + 4 = 5.$$

$$B = -2.$$

$$A = 3, B = -2, C = 4.$$

3

(b) $x + y = -1$ 

$$3x + 2y = 0. \quad \textcircled{A}$$

$$3x + 3y = -3. \quad \textcircled{B}$$

$$\textcircled{B} - \textcircled{A}$$

$$y = -3.$$

Sub into
1st.

$$x = 2.$$

3

$$(c) \quad y = a(x-4)x. \quad (2, -8)$$

$$-8 = a(-2)2.$$

$$a = 2.$$

$$y = 2x(x-4)$$

$$y = 2x^2 - 8x. \quad 2$$

$$(d) \quad y = -2x + 5$$

$$m_1 = -2$$

$$m_1 \times m_2 = -1$$

$$m_2 = \frac{1}{2}. \quad (5, 1)$$

$$y - 1 = \frac{1}{2}(x - 5).$$

$$2y - 2 = x - 5$$

$$x - 2y - 3 = 0$$

$$Q15 (a) (i) \alpha + \beta = 2$$

$$(ii) \alpha\beta = -4$$

$$(iii) \frac{\alpha\beta}{\alpha+\beta} = \frac{-4}{2} = -\frac{1}{2} \quad 3$$

$$(iv) (\alpha+\beta)^2 - 2\alpha\beta = 4 + 8 = 12 \quad 2$$

$$(b) \quad x^2 - (k+5)x + 9 = 0$$

$$(i) \quad \Delta = 0$$

$$\Delta = (k+5)^2 - 4 \times 9$$

$$= k^2 + 10k + 25 - 36$$

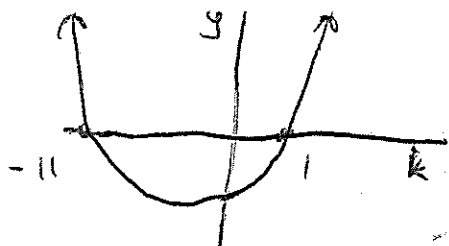
$$= k^2 + 10k - 11$$

$$k^2 + 10k - 11 = 0.$$

$$(k+11)(k-1) = 0.$$

$$k = -11, 1.$$

$$(ii) \quad \Delta < 0.$$



$$-11 < k < 1$$

$$(c) \quad (x - 2 - \sqrt{3})(x - 2 + \sqrt{3}) = 0.$$

$$(x - (2 + \sqrt{3}))(x - (2 - \sqrt{3})) = 0.$$

$$x^2 - 4x + (2 + \sqrt{3})(2 - \sqrt{3}) = 0$$

$$x^2 - 4x + 4 - 3 = 0$$

$$x^2 - 4x + 1 = 0.$$

$$(d) (i) P = (60 - \frac{1}{2}y) - (\frac{1}{3}y^2 + 45y + 27)$$

$$Profit = 33 - \frac{91}{2}y - \frac{1}{3}y^2.$$

$$(ii) y = \frac{\frac{91}{2}}{-\frac{2}{3}}$$

4

$$= -\frac{273}{4}.$$

$$\text{Max Profit} = \$1585.69.$$